Jiahong Li

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Education and Training

Postdoc in Medical Engineering — Jun. 2026 -

- California Institute of Technology, Pasadena, CA
- Advisor: Wei Gao

Ph.D. in Medical and Electrical Engineering — Sep. 2020 - Jun. 2026

- California Institute of Technology, Pasadena, CA
- Advisor: Wei Gao

M.S. in Materials Science and Engineering — Sep. 2018 - Dec. 2019

- Northwestern University, Evanston, IL
- Advisor: John Rogers

B.S. in Chemistry — Sep. 2014 – Jun. 2018

- Wuhan University, Wuhan, China

Publication

- **Jiahong Li**, Kexin Fan, Xiaotian Ma, Canran Wang, Jihong Min, Yonglin Chen, Yu Song, Yadong Xu, Wei Gao*. "Strain-insensitive wet-tissue-adhesive biphasic bioelectronics for physicochemical monitoring and adaptive therapy" *Nature Materials* (**2026**) [Link]
- **Jiahong Li**, Jin Qu, Wei Gao*. "Tissue-bioelectronics interface." *Chemical Society Reviews* (**2026**) [Link]
- Songsong Tang, Hong Han, Xiaotian Ma, Payal N Patel, Chen Gong, Junhang Zhang, Ernesto Criado-Hidalgo, Jounghyun Yoo, **Jiahong Li**, Gwangmook Kim, Shukun Yin, Di Wu, Mikhail G Shapiro, Qifa Zhou, Wei Gao*. "Enzymatic microbubble robots." *Nature Nanotechnology* (**2026**) [Link]
- Elham Davoodi, **Jiahong Li**, Xiaotian Ma, Alireza Hasani Najafabadi, Jounghyun Yoo, Gengxi Lu, Ehsan Shirzaei Sani, Sunho Lee, Hossein Montazerian, Gwangmook Kim, Jason Williams, Jee Won Yang, Yushun Zeng, Lei S Li, Zhiyang Jin, Behnam Sadri, Shervin S Nia, Lihong V Wang, Tzung K Hsiai, Paul S Weiss, Qifa Zhou, Ali Khademhosseini, Di Wu, Mikhail G Shapiro, Wei Gao*. "Imaging-guided deep tissue in vivo sound printing." *Science* (**2025**) [Link]
- Canran Wang†, Kexin Fan†, Ehsan Shirzaei Sani, José A Lasalde-Ramírez, Wenzheng Heng, Jihong Min, Samuel A Solomon, Minqiang Wang, **Jiahong Li**, Hong Han, Gwangmook Kim, Soyoung Shin, Alex Seder, Chia-Ding Shih, David G Armstrong, Wei Gao*. "A microfluidic wearable device for wound exudate management and analysis in human chronic wounds" *Science Translational Medicine* (**2025**) [Link]
- Hossein Montazerian†, Elham Davoodi†, Canran Wang, Farnaz Lorestani, **Jiahong Li**,

Reihaneh Haghniaz, Rohan R Sampath, Neda Mohaghegh, Safoora Khosravi, Fatemeh Zehtabi, Yichao Zhao, Negar Hosseinzadeh, Tianhan Liu, Tzung K Hsiai, Alireza Hassani Najafabadi*, Robert Langer, Daniel G Anderson, Paul S Weiss*, Ali Khademhosseini*, Wei Gao*. “Boosting hydrogel conductivity via water-dispersible conducting polymers for injectable bioelectronics.” *Nature Communications* (**2025**) [Link]

- Wenzheng Heng, Shukun Yin, Jihong Min, Canran Wang, Hong Han, Ehsan Shirzaei Sani, **Jiahong Li**, Yu Song, Harry B Rossiter, Wei Gao*. “A smart mask for exhaled breath condensate harvesting and analysis” *Science* (**2024**) [Link]
- Yadong Xu, Zhilu Ye, Ganggang Zhao, Qihui Fei, Zehua Chen, **Jiahong Li**, Minye Yang, Yichong Ren, Benton Berigan, Yun Ling, Xiaoyan Qian, Lin Shi, Ilker Ozden, Jingwei Xie, Wei Gao*, Pai-Yen Chen*, Zheng Yan*. “Phase-separated porous nanocomposite with ultralow percolation threshold for wireless bioelectronics.” *Nature Nanotechnology* (**2024**) [Link]
- Changhao Xu†, Yu Song†, Juliane R Sempionatto†, Samuel A Solomon†, You Yu, Hnin YY Nyein, Roland Yingjie Tay, **Jiahong Li**, Wenzheng Heng, Jihong Min, Alison Lao, Tzung K Hsiai, Jennifer A Sumner, Wei Gao*. “A physicochemical-sensing electronic skin for stress response monitoring.” *Nature Electronics* (**2024**) [Link]
- **Jiahong Li**, Yadong Xu, Wei Gao* “Injectable and retrievable soft electronics.” *Nature Materials* (**2024**) [Link]
- Cui Ye†, Minqiang Wang†, Jihong Min, Roland Yingjie Tay, Heather Lukas, Juliane R Sempionatto, **Jiahong Li**, Changhao Xu, Wei Gao*. “A wearable aptamer nanobiosensor for non-invasive female hormone monitoring.” *Nature Nanotechnology* (**2023**) [Link]
- Yu Song†, Roland Yingjie Tay†, **Jiahong Li**, Changhao Xu, Jihong Min, Ehsan Shirzaei Sani, Gwangmook Kim, Wenzheng Heng, Inho Kim, Wei Gao*. “3D-printed epifluidic electronic skin for machine learning–powered multimodal health surveillance.” *Science Advances* (**2023**) [Link]
- Jiaobing Tu, Jihong Min, Yu Song, Changhao Xu, Jiahong Li, Jeff Moore, Justin Hanson, Erin Hu, Tanyalak Parimon, Ting-Yu Wang, Elham Davoodi, Tsui-Fen Chou, Peter Chen, Jeffrey J. Hsu. Harry B Rossiter, Wei Gao*. "A wireless patch for the monitoring of C-reactive protein in sweat." *Nature Biomedical Engineering* (**2023**) [Link]
- Shirzaei Sani Ehsan†, Changhao Xu†, Canran Wang, Yu Song, Jihong Min, Jiaobing Tu, Samuel A. Solomon, **Jiahong Li**, Jaminelli L. Banks, David G. Armsterong, Wei Gao*. "A stretchable wireless wearable bioelectronic system for multiplexed monitoring and combination treatment of infected chronic wounds." *Science Advances* (**2023**) [Link]
- Yu You†, **Jiahong Li†,** Samuel A. Solomon†, Jihong Min, Jiaobing Tu, Wei Guo, Changhao Xu, Yu Song, and Wei Gao*. "All-printed soft human-machine interface for robotic physicochemical sensing." *Science robotics* (**2022**) [Link]
- Zhaoqing Cong, Songsong Tang*, Leiming Xie, Ming Yang, Yangyang Li, Dongdong Lu, **Jiahong Li**, Qingxin Yang, Qiwei Chen, Zhiqiang Zhang, Xueji Zhang, Song Wu*. "Magnetic-Powered Janus Cell Robots Loaded with Oncolytic Adenovirus for Active and

Targeted Virotherapy of Bladder Cancer." *Advanced Materials* (**2022**) [Link]

- Xiaogang Guo†, Xiaoyue Ni†, **Jiahong Li†**, Hang Zhang, Fan Zhang, Huabin Yu, Jun Wu, Hongshuai Lei, Yonggang Huang, John A. Rogers*, Yihui Zhang*. "Designing mechanical metamaterials with kirigami-inspired, hierarchical constructions for giant positive and negative thermal expansion." *Advanced Materials* (**2020**) [Link]
- Xiaoyue Ni†, Xiaogang Guo†, **Jiahong Li**, Yonggang Huang, Yihui Zhang*, John A. Rogers*. "Two-Dimensional Mechanical Metamaterials with Widely-Tunable Unusual Modes of Thermal Expansion." *Advanced Materials.* (**2019**). [Link]
- Jinfeng Cao†**, Jiahong Li†,** Yumei Chen, Lina Zhang, and Jinping Zhou*. "Dual Physical Crosslinking Strategy to Construct Moldable Hydrogels with Ultrahigh Strength and Toughness." *Advanced Functional Materials* (**2018**) [Link]

Teaching

- Sensors in Medicine. MedE 202, Caltech. Winter 2025.

Academic Service

- Guest Editor for Sensors, Special issue on Smart Wearable Biosensors for Healthcare Monitoring, 2026
- Independent Reviewer of the Following Journals:
  IEEE Flexible Electronics, Transactions on Biomedical Engineering, Scientific Reports, Sensors and Diagnostics, Materials Today Chemistry, Biosensors and Bioelectronics, Discover Chemistry, npj Flexible Electronics, Health Nanotechnology, Microsystems and Nanoengineering

Award and Honors

- Rising Star in Materials Science & Engineering, MIT, 2026
- Dr. David C. Gakenheimer Fellow, Caltech, 2024
- Andrew and Peggy Cherng Fellowship, Caltech, 2020
- Outstanding Graduation Thesis, Wuhan University 2018
- Samsung Fellowship, Wuhan University, 2016
- Outstanding Student Scholarship, Wuhan University, 2015